Session-5-exercises.R

```
##########################
### Session 5 Exercises ###
##########################

#NA, NaN, nanana!
# What happens if I divide a number by 0

numbirds <- c(1,2,3,4/0,0/0,NA)
numbirds

## [1]   1   2   3 Inf NaN  NA

is.na(numbirds)

## [1] FALSE FALSE FALSE FALSE  TRUE  TRUE

is.nan(numbirds)

## [1] FALSE FALSE FALSE FALSE  TRUE FALSE

is.infinite(numbirds)

## [1] FALSE FALSE FALSE  TRUE FALSE FALSE

is.finite(numbirds)

## [1]  TRUE  TRUE  TRUE FALSE FALSE FALSE

#Filter missing
library(tidyverse)

names <- c("ID", "DBAName", "AKAName", "License", "FacilityType", "Risk", "Address",
           "City", "State", "ZIP", "InspectionDate", "InspectionType", "Results",
           "Violations", "Latitude","Longitude","Location")

inspections <- read_csv('inspections.csv',
                        col_names=names, skip=1)

# Look at a summary of the data
summary(inspections)

##        ID            DBAName             AKAName             License
##  Min.   :  44247   Length:145606      Length:145606      Min.   :      0
##  1st Qu.: 634510   Class :character   Class :character   1st Qu.:1120535
##  Median :1335522   Mode  :character   Mode  :character   Median :1904431
##  Mean   :1193936                                         Mean   :1501224
##  3rd Qu.:1537693                                         3rd Qu.:2141324
##  Max.   :2014211                                         Max.   :9999999
##                                                          NA's   :14
```

```
##  FacilityType          Risk             Address            City          
##  Length:145606      Length:145606      Length:145606      Length:145606     
##  Class :character   Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character  
##                                                                             
##                                                                             
##                                                                             
##                                                                             
##     State                ZIP        InspectionDate     InspectionType    
##  Length:145606      Min.   :60007   Length:145606      Length:145606     
##  Class :character   1st Qu.:60614   Class :character   Class :character  
##  Mode  :character   Median :60625   Mode  :character   Mode  :character  
##                     Mean   :60629                                        
##                     3rd Qu.:60643                                        
##                     Max.   :60827                                        
##                     NA's   :97                                           
##    Results           Violations           Latitude       Longitude     
##  Length:145606      Length:145606      Min.   :41.64   Min.   :-87.91  
##  Class :character   Class :character   1st Qu.:41.83   1st Qu.:-87.71  
##  Mode  :character   Mode  :character   Median :41.89   Median :-87.67  
##                                        Mean   :41.88   Mean   :-87.68  
##                                        3rd Qu.:41.94   3rd Qu.:-87.63  
##                                        Max.   :42.02   Max.   :-87.53  
##                                        NA's   :513     NA's   :513     
##    Location        
##  Length:145606     
##  Class :character  
##  Mode  :character  
##                    
##                    
##                    
##                    
```

```r
# Which inspections have NA values for license?
unlicensed <- inspections %>%
  filter(is.na(License))

licensed <- inspections %>%
  filter(!is.na(License))

#Solving the mystery of missing zeros. It will be hot.
# Load the data file
heating <- read_csv("heating.csv")
```

```
message.
```

```r
glimpse(heating)
```

```
## Rows: 14
## Columns: 9
## $ Source                   <chr> "Warm-air furnace", "Steam or hot water system…
## $ `Under 25 years old`     <chr> "2546", "326", "529", "280", "267", "15", "18"…
## $ `25 to 29 years old`     <chr> "5061", "672", "919", "430", "423", "29", "38"…
## $ `30 to 34 years old`     <chr> "6701", "926", "1106", "471", "513", "66", "42…
```

```
## $ `35 to 44 years old`    <dbl> 13609, 1718, 2288, 680, 1153, 120, 57, 276, 22…
## $ `45 to 54 years old`    <dbl> 15348, 2137, 2733, 776, 1188, 176, 139, 339, 2…
## $ `55 to 64 years old`    <dbl> 15172, 2325, 2848, 901, 1023, 153, 166, 436, 3…
## $ `65 to 74 years old`    <dbl> 10380, 1556, 2010, 542, 790, 104, 99, 261, 205…
## $ `75 years old and over` <dbl> 7389, 1260, 1546, 535, 606, 113, 89, 166, 80, …
```

```
# Tidy the data
heating <- heating %>%
  pivot_longer(-Source, names_to ="age", values_to="homes",
               values_transform = list(homes = as.character))

# Look at a summary
glimpse(heating)
```

```
## Rows: 112
## Columns: 3
## $ Source <chr> "Warm-air furnace", "Warm-air furnace", "Warm-air furnace", "Wa…
## $ age    <chr> "Under 25 years old", "25 to 29 years old", "30 to 34 years old…
## $ homes  <chr> "2546", "5061", "6701", "13609", "15348", "15172", "10380", "73…
```

```
summary(heating)
```

```
##     Source              age                homes          
##  Length:112         Length:112         Length:112        
##  Class :character   Class :character   Class :character  
##  Mode  :character   Mode  :character   Mode  :character  
```

```
# Attempt to convert homes from character to numeric
heating %>%
  mutate(homes=as.numeric(homes))
```

```
## Warning: There was 1 warning in `mutate()`.
## ℹ In argument: `homes = as.numeric(homes)`.
## Caused by warning:
## ! NAs introduced by coercion
```

```
## # A tibble: 112 × 3
##    Source                   age                   homes
##    <chr>                    <chr>                 <dbl>
##  1 Warm-air furnace         Under 25 years old     2546
##  2 Warm-air furnace         25 to 29 years old     5061
##  3 Warm-air furnace         30 to 34 years old     6701
##  4 Warm-air furnace         35 to 44 years old    13609
##  5 Warm-air furnace         45 to 54 years old    15348
##  6 Warm-air furnace         55 to 64 years old    15172
##  7 Warm-air furnace         65 to 74 years old    10380
##  8 Warm-air furnace         75 years old and over  7389
##  9 Steam or hot water system Under 25 years old      326
## 10 Steam or hot water system 25 to 29 years old      672
## # ℹ 102 more rows
```

```
# Find those values
heating %>%
  filter(is.na(as.numeric(homes)))
```

```
## Warning: There was 1 warning in `filter()`.
## ℹ In argument: `is.na(as.numeric(homes))`.
## Caused by warning:
## ! NAs introduced by coercion

## # A tibble: 3 × 3
##   Source        age                homes
##   <chr>         <chr>              <chr>
## 1 Cooking stove Under 25 years old .
## 2 Cooking stove 25 to 29 years old Z
## 3 Cooking stove 30 to 34 years old .

# Replace with zeros
heating %>%
  mutate(homes=ifelse(homes==".", 0, homes)) %>%
  mutate(homes=ifelse(homes=="Z", 0, homes)) %>%
  mutate(homes=as.numeric(homes)) %>%
  filter(Source=="Cooking stove")

## # A tibble: 8 × 3
##   Source        age                      homes
##   <chr>         <chr>                    <dbl>
## 1 Cooking stove Under 25 years old           0
## 2 Cooking stove 25 to 29 years old           0
## 3 Cooking stove 30 to 34 years old           0
## 4 Cooking stove 35 to 44 years old          22
## 5 Cooking stove 45 to 54 years old          20
## 6 Cooking stove 55 to 64 years old          11
## 7 Cooking stove 65 to 74 years old          11
## 8 Cooking stove 75 years old and over        3

#Apply on the dataset
heating <- heating %>%
  mutate(homes=ifelse(homes==".", 0, homes)) %>%
  mutate(homes=ifelse(homes=="Z", 0, homes)) %>%
  mutate(homes=as.numeric(homes))

#When nothing is missing... Doch, vermisste Schizophrene sind nicht allein

# Load the data file
land <- read_csv("publiclands.csv")

# Look at the data
summary(land)

##     State           PublicLandAcres
##  Length:42          Min.   :   16000
##  Class :character   1st Qu.:  606250
##  Mode  :character   Median : 1156000
##                     Mean   : 4577905
##                     3rd Qu.: 7592500
##                     Max.   :22083000
```

```r
# How many rows are there?
nrow(land)
## [1] 42

unique(land$State)

##  [1] "Alabama"        "Alaska"         "Arizona"        "Arkansas"
##  [5] "California"     "Colorado"       "Florida"        "Georgia"
##  [9] "Idaho"          "Illinois"       "Indiana"        "Kansas"
## [13] "Kentucky"       "Louisiana"      "Maine"          "Michigan"
## [17] "Minnesota"      "Mississippi"    "Missouri"       "Montana"
## [21] "Nebraska"       "Nevada"         "New Hampshire"  "New Mexico"
## [25] "New York"       "North Carolina" "North Dakota"   "Ohio"
## [29] "Oklahoma"       "Oregon"         "Pennsylvania"   "South Carolina"
## [33] "South Dakota"   "Tennessee"      "Texas"          "Utah"
## [37] "Vermont"        "Virginia"       "Washington"     "West Virginia"
## [41] "Wisconsin"      "Wyoming"

# Insert missing states
missing_states <- tibble(State=c('Connecticut', 'Delaware', 'Hawaii', 'Iowa',
                    'Maryland', 'Massachusetts','New Jersey', 'Rhode Island'),
                         PublicLandAcres=c(0,0,0,0,0,0,0,0))

land_full <- rbind(land, missing_states)


#Alternatively
land_full <- bind_rows(land, missing_states)
View(land_full)

mean(land$PublicLandAcres)
## [1] 4577905

mean(land_full$PublicLandAcres)
## [1] 3845440

#Almost ready for ML 101 course

# Load the data file
iris <- read_csv("dirty petal.csv")

# Look at the data
View(iris)

# What happens when we try some aggregate functions?
sum(iris$Petal.Length)
## [1] NA

mean(iris$Petal.Length)
## [1] NA

max(iris$Petal.Length)
## [1] NA
```

```r
# We can remove missing values before our calculation
sum(iris$Petal.Length, na.rm=TRUE)
## [1] 582.945

mean(iris$Petal.Length, na.rm=TRUE)
## [1] 4.449962

max(iris$Petal.Length, na.rm=TRUE)
## [1] 63

#Olympics forever
# Load the data file
olympics <- read_csv("olympic games.txt")

# Remove duplicated rows
olympics <- unique(olympics)

# Filter out illogical records for Japan
olympics <- olympics %>%
  filter(!(City=='Tokyo, Japan' & Year==2020))

# View the dataset again
View(olympics)


# Pop (on the rock)
# Load the data file
library(readxl)
france <- read_excel("France population 2015.xlsx")

## New names:
## • `` -> `...2`
## • `` -> `...3`
## • `` -> `...4`

#Focus on the right range:
france <- read_excel("France population 2015.xlsx", range = "C4:D12")

# Take a look
glimpse(france)

## Rows: 8
## Columns: 2
## $ Category   <chr> "Male", "Female", "0-14 years", "15-24 years", "25-54 years…
## $ Population <dbl> 32459000, 34216000, 12231000, 7978000, 26704000, 8238000, 1…

# What is the population of France?
sum(france$Population)
## [1] 266700000
```

```r
# Remove total and gender breakouts
france <- france %>%
  filter(!(Category %in% c('Total', 'Male', 'Female')))

# What is the population of France?
sum(france$Population)
```

```
## [1] 66675000
```

```r
#Mexican weather cleaner
# Load the data file
weather <- read_csv("mexicanweather.csv")

# Let's look at what we have
weather
```

```
## # A tibble: 33,712 × 4
##    station     element value date
##    <chr>       <chr>   <dbl> <date>
##  1 MX000017004 TMAX      310 1955-04-01
##  2 MX000017004 TMIN      150 1955-04-01
##  3 MX000017004 TMAX      310 1955-05-01
##  4 MX000017004 TMIN      200 1955-05-01
##  5 MX000017004 TMAX      300 1955-06-01
##  6 MX000017004 TMIN      160 1955-06-01
##  7 MX000017004 TMAX      270 1955-07-01
##  8 MX000017004 TMIN      150 1955-07-01
##  9 MX000017004 TMAX      230 1955-08-01
## 10 MX000017004 TMIN      140 1955-08-01
## # ℹ 33,702 more rows
```

```r
# And use pivot_wider() to make it wider
weather<- pivot_wider(weather, names_from=element, values_from=value)

# Where are we now?
weather
```

```
## # A tibble: 16,871 × 4
##    station     date        TMAX  TMIN
##    <chr>       <date>     <dbl> <dbl>
##  1 MX000017004 1955-04-01   310   150
##  2 MX000017004 1955-05-01   310   200
##  3 MX000017004 1955-06-01   300   160
##  4 MX000017004 1955-07-01   270   150
##  5 MX000017004 1955-08-01   230   140
##  6 MX000017004 1955-09-01   230   150
##  7 MX000017004 1955-10-01   240   150
##  8 MX000017004 1955-11-01   270   130
##  9 MX000017004 1955-12-01   240   130
## 10 MX000017004 1956-01-01   240    90
## # ℹ 16,861 more rows
```

```r
# That's the right format, but take a look at the data

tail(weather)

## # A tibble: 6 × 4
##   station     date        TMAX  TMIN
##   <chr>       <date>     <dbl> <dbl>
## 1 MX000017004 2010-10-30    NA    NA
## 2 MX000017004 2010-11-30    NA    NA
## 3 MX000017004 2010-12-30    NA    NA
## 4 MX000017004 2011-01-30    NA    NA
## 5 MX000017004 2011-03-30    NA    NA
## 6 MX000017004 2011-04-30    NA    NA

# It's pretty sparse.  We really don't need all of those lines that have two
NA values
weather <- weather %>%
  filter(!(is.na(TMAX) & is.na(TMIN)))

# Let's make those column names nicer
# And just for tidyness, let's put the min before the max
weather <- weather %>%
  rename(maxtemp=TMAX, mintemp=TMIN) %>%
  select(station, date, mintemp, maxtemp)

head(weather, n=20)

## # A tibble: 20 × 4
##    station     date       mintemp maxtemp
##    <chr>       <date>       <dbl>   <dbl>
##  1 MX000017004 1955-04-01     150     310
##  2 MX000017004 1955-05-01     200     310
##  3 MX000017004 1955-06-01     160     300
##  4 MX000017004 1955-07-01     150     270
##  5 MX000017004 1955-08-01     140     230
##  6 MX000017004 1955-09-01     150     230
##  7 MX000017004 1955-10-01     150     240
##  8 MX000017004 1955-11-01     130     270
##  9 MX000017004 1955-12-01     130     240
## 10 MX000017004 1956-01-01      90     240
## 11 MX000017004 1956-02-01     120     260
## 12 MX000017004 1956-03-01     130     290
## 13 MX000017004 1956-04-01     160     320
## 14 MX000017004 1956-05-01     140     280
## 15 MX000017004 1956-06-01     140     230
## 16 MX000017004 1956-07-01     140     250
## 17 MX000017004 1956-08-01     140     250
## 18 MX000017004 1956-09-01     150     270
## 19 MX000017004 1956-10-01     160     260
## 20 MX000017004 1956-12-01     120     240
```

```
# First, divide the temperatures by 10 to get them in degrees Celsius
weather <- weather %>%
  mutate(mintemp = mintemp/10) %>%
  mutate(maxtemp = maxtemp/10)

# Next, convert them to Fahrenheit
weather_fahrenheit <- weather %>%
  mutate(mintemp=mintemp*(9/5)+32) %>%
  mutate(maxtemp=maxtemp*(9/5)+32)

#Exercise 5.1
#a
coffee <- read_csv("coffeeshop.csv")

## New names:
## Rows: 17 Columns: 243
## ── Column specification
## ──────────────────────────────────────────────────────── Delimiter: "," chr
## (243): Beverage_category, Coffee...2, Coffee...3, Coffee...4, Coffee...5...
## ℹ Use `spec()` to retrieve the full column specification for this data. ℹ
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## • `Coffee` -> `Coffee...2`
## • `Coffee` -> `Coffee...3`
## • `Coffee` -> `Coffee...4`
## • `Coffee` -> `Coffee...5`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...6`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...7`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...8`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...9`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...10`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...11`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...12`
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## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...14`
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## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...33`
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```

```
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...35`
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## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...62`
## • `Classic Espresso Drinks` -> `Classic Espresso Drinks...63`
## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...64`
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## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...89`
```

```
## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...90`
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## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...101`
## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...102`
## • `Signature Espresso Drinks` -> `Signature Espresso Drinks...103`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...104`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...105`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...107`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...120`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...121`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...122`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...123`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...125`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...144`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...145`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...146`
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## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...154`
## • `Tazo Tea Drinks` -> `Tazo Tea Drinks...155`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...156`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...157`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...158`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...159`
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## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...165`
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## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...167`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...168`
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## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...170`
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## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...172`
## • `Shaken Iced Beverages` -> `Shaken Iced Beverages...173`
## • `Smoothies` -> `Smoothies...174`
## • `Smoothies` -> `Smoothies...175`
## • `Smoothies` -> `Smoothies...176`
## • `Smoothies` -> `Smoothies...177`
## • `Smoothies` -> `Smoothies...178`
## • `Smoothies` -> `Smoothies...179`
## • `Smoothies` -> `Smoothies...180`
## • `Smoothies` -> `Smoothies...181`
## • `Smoothies` -> `Smoothies...182`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...183`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...184`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...185`
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## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...197`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...198`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...199`
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## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...200`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...201`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...202`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...203`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...204`
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## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...206`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...207`
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## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...214`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...215`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...216`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...217`
## • `Frappuccino Blended Coffee` -> `Frappuccino Blended Coffee...218`
## • `Frappuccino Light Blended Coffee` -> `Frappuccino Light Blended
##   Coffee...219`
## • `Frappuccino Light Blended Coffee` -> `Frappuccino Light Blended
##   Coffee...220`
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##   Coffee...221`
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##   Coffee...223`
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##   Coffee...224`
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##   Coffee...225`
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##   Coffee...227`
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##   Coffee...228`
## • `Frappuccino Light Blended Coffee` -> `Frappuccino Light Blended
##   Coffee...229`
## • `Frappuccino Light Blended Coffee` -> `Frappuccino Light Blended
##   Coffee...230`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...231`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...232`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...233`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...234`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...235`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...236`
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## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...239`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...240`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...241`
```

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## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...242`
## • `Frappuccino Blended Creme` -> `Frappuccino Blended Creme...243`
```

```
glimpse(coffee)
```

```
## Rows: 17
## Columns: 243
## $ Beverage_category                          <chr> "Beverage", "Beverage_prep", …
## $ Coffee...2                                 <chr> "Brewed Coffee", "Short", "3"…
## $ Coffee...3                                 <chr> "Brewed Coffee", "Tall", "4",…
## $ Coffee...4                                 <chr> "Brewed Coffee", "Grande", "5…
## $ Coffee...5                                 <chr> "Brewed Coffee", "Venti", "5"…
## $ `Classic Espresso Drinks...6`              <chr> "Caffe Latte", "Short Nonfat …
## $ `Classic Espresso Drinks...7`              <chr> "Caffe Latte", "2% Milk", "10…
## $ `Classic Espresso Drinks...8`              <chr> "Caffe Latte", "Soymilk", "70…
## $ `Classic Espresso Drinks...9`              <chr> "Caffe Latte", "Tall Nonfat M…
## $ `Classic Espresso Drinks...10`             <chr> "Caffe Latte", "2% Milk", "15…
## $ `Classic Espresso Drinks...11`             <chr> "Caffe Latte", "Soymilk", "11…
## $ `Classic Espresso Drinks...12`             <chr> "Caffe Latte", "Grande Nonfat…
## $ `Classic Espresso Drinks...13`             <chr> "Caffe Latte", "2% Milk", "19…
## $ `Classic Espresso Drinks...14`             <chr> "Caffe Latte", "Soymilk", "15…
## $ `Classic Espresso Drinks...15`             <chr> "Caffe Latte", "Venti Nonfat …
## $ `Classic Espresso Drinks...16`             <chr> "Caffe Latte", "2% Milk", "24…
## $ `Classic Espresso Drinks...17`             <chr> "Caffe Latte", "Soymilk", "19…
## $ `Classic Espresso Drinks...18`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...19`             <chr> "Caffe Mocha (Without Whipped…
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## $ `Classic Espresso Drinks...21`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...22`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...23`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...24`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...25`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...26`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...27`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...28`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...29`             <chr> "Caffe Mocha (Without Whipped…
## $ `Classic Espresso Drinks...30`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...31`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...32`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...33`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...34`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...35`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...36`             <chr> "Vanilla Latte (Or Other Flav…
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## $ `Classic Espresso Drinks...39`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...40`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...41`             <chr> "Vanilla Latte (Or Other Flav…
## $ `Classic Espresso Drinks...42`             <chr> "Caffe Americano", "Short", "…
## $ `Classic Espresso Drinks...43`             <chr> "Caffe Americano", "Tall", "1…
## $ `Classic Espresso Drinks...44`             <chr> "Caffe Americano", "Grande", …
## $ `Classic Espresso Drinks...45`             <chr> "Caffe Americano", "Venti", "…
## $ `Classic Espresso Drinks...46`             <chr> "Cappuccino", "Short Nonfat M…
## $ `Classic Espresso Drinks...47`             <chr> "Cappuccino", "2% Milk", "80"…
## $ `Classic Espresso Drinks...48`             <chr> "Cappuccino", "Soymilk", "50"…
```

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## $ `Classic Espresso Drinks...49`          <chr> "Cappuccino", "Tall Nonfat Mi…
## $ `Classic Espresso Drinks...50`          <chr> "Cappuccino", "2% Milk", "90"…
## $ `Classic Espresso Drinks...51`          <chr> "Cappuccino", "Soymilk", "70"…
## $ `Classic Espresso Drinks...52`          <chr> "Cappuccino", "Grande Nonfat …
## $ `Classic Espresso Drinks...53`          <chr> "Cappuccino", "2% Milk", "120…
## $ `Classic Espresso Drinks...54`          <chr> "Cappuccino", "Soymilk", "100…
## $ `Classic Espresso Drinks...55`          <chr> "Cappuccino", "Venti Nonfat M…
## $ `Classic Espresso Drinks...56`          <chr> "Cappuccino", "2% Milk", "150…
## $ `Classic Espresso Drinks...57`          <chr> "Cappuccino", "Soymilk", "120…
## $ `Classic Espresso Drinks...58`          <chr> "Espresso", "Solo", "5", "0",…
## $ `Classic Espresso Drinks...59`          <chr> "Espresso", "Doppio", "10", "…
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## $ `Classic Espresso Drinks...63`          <chr> "Skinny Latte (Any Flavour)",…
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## $ `Signature Espresso Drinks...65`        <chr> "Caramel Macchiato", "2% Milk…
## $ `Signature Espresso Drinks...66`        <chr> "Caramel Macchiato", "Soymilk…
## $ `Signature Espresso Drinks...67`        <chr> "Caramel Macchiato", "Tall No…
## $ `Signature Espresso Drinks...68`        <chr> "Caramel Macchiato", "2% Milk…
## $ `Signature Espresso Drinks...69`        <chr> "Caramel Macchiato", "Soymilk…
## $ `Signature Espresso Drinks...70`        <chr> "Caramel Macchiato", "Grande …
## $ `Signature Espresso Drinks...71`        <chr> "Caramel Macchiato", "2% Milk…
## $ `Signature Espresso Drinks...72`        <chr> "Caramel Macchiato", "Soymilk…
## $ `Signature Espresso Drinks...73`        <chr> "Caramel Macchiato", "Venti N…
## $ `Signature Espresso Drinks...74`        <chr> "Caramel Macchiato", "2% Milk…
## $ `Signature Espresso Drinks...75`        <chr> "Caramel Macchiato", "Soymilk…
## $ `Signature Espresso Drinks...76`        <chr> "White Chocolate Mocha (Witho…
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## $ `Signature Espresso Drinks...103`       <chr> "Caramel Apple Spice (Without…
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## $ `Tazo Tea Drinks...106`                    <chr> "Tazo Tea", "Grande", "0", "0…
## $ `Tazo Tea Drinks...107`                    <chr> "Tazo Tea", "Venti", "0", "0"…
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## $ `Tazo Tea Drinks...109`                    <chr> "Tazo Chai Tea Latte", "2% Mi…
## $ `Tazo Tea Drinks...110`                    <chr> "Tazo Chai Tea Latte", "Soymi…
## $ `Tazo Tea Drinks...111`                    <chr> "Tazo Chai Tea Latte", "Tall …
## $ `Tazo Tea Drinks...112`                    <chr> "Tazo Chai Tea Latte", "2% Mi…
## $ `Tazo Tea Drinks...113`                    <chr> "Tazo Chai Tea Latte", "Soymi…
## $ `Tazo Tea Drinks...114`                    <chr> "Tazo Chai Tea Latte", "Grand…
## $ `Tazo Tea Drinks...115`                    <chr> "Tazo Chai Tea Latte", "2% Mi…
## $ `Tazo Tea Drinks...116`                    <chr> "Tazo Chai Tea Latte", "Soymi…
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## $ `Tazo Tea Drinks...118`                    <chr> "Tazo Chai Tea Latte", "2% Mi…
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## $ `Tazo Tea Drinks...121`                    <chr> "Tazo Green Tea Latte", "2% M…
## $ `Tazo Tea Drinks...122`                    <chr> "Tazo Green Tea Latte", "Soym…
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## $ `Tazo Tea Drinks...134`                    <chr> "Tazo Full-Leaf Tea Latte", "…
## $ `Tazo Tea Drinks...135`                    <chr> "Tazo Full-Leaf Tea Latte", "…
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## $ `Tazo Tea Drinks...137`                    <chr> "Tazo Full-Leaf Tea Latte", "…
## $ `Tazo Tea Drinks...138`                    <chr> "Tazo Full-Leaf Tea Latte", "…
## $ `Tazo Tea Drinks...139`                    <chr> "Tazo Full-Leaf Tea Latte", "…
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## $ `Shaken Iced Beverages...158`              <chr> "Iced Brewed Coffee (With Cla…
```

```
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## $ `Shaken Iced Beverages...170`          <chr> "Shaken Iced Tazo Tea (With C…
## $ `Shaken Iced Beverages...171`          <chr> "Shaken Iced Tazo Tea Lemonad…
## $ `Shaken Iced Beverages...172`          <chr> "Shaken Iced Tazo Tea Lemonad…
## $ `Shaken Iced Beverages...173`          <chr> "Shaken Iced Tazo Tea Lemonad…
## $ Smoothies...174                        <chr> "Banana Chocolate Smoothie", …
## $ Smoothies...175                        <chr> "Banana Chocolate Smoothie", …
## $ Smoothies...176                        <chr> "Banana Chocolate Smoothie", …
## $ Smoothies...177                        <chr> "Orange Mango Banana Smoothie…
## $ Smoothies...178                        <chr> "Orange Mango Banana Smoothie…
## $ Smoothies...179                        <chr> "Orange Mango Banana Smoothie…
## $ Smoothies...180                        <chr> "Strawberry Banana Smoothie",…
## $ Smoothies...181                        <chr> "Strawberry Banana Smoothie",…
## $ Smoothies...182                        <chr> "Strawberry Banana Smoothie",…
## $ `Frappuccino Blended Coffee...183`     <chr> "Coffee", "Tall Nonfat Milk",…
## $ `Frappuccino Blended Coffee...184`     <chr> "Coffee", "Whole Milk", "180"…
## $ `Frappuccino Blended Coffee...185`     <chr> "Coffee", "Soymilk", "160", "…
## $ `Frappuccino Blended Coffee...186`     <chr> "Coffee", "Grande Nonfat Milk…
## $ `Frappuccino Blended Coffee...187`     <chr> "Coffee", "Whole Milk", "240"…
## $ `Frappuccino Blended Coffee...188`     <chr> "Coffee", "Soymilk", "220", "…
## $ `Frappuccino Blended Coffee...189`     <chr> "Coffee", "Venti Nonfat Milk"…
## $ `Frappuccino Blended Coffee...190`     <chr> "Coffee", "Whole Milk", "350"…
## $ `Frappuccino Blended Coffee...191`     <chr> "Coffee", "Soymilk", "310", "…
## $ `Frappuccino Blended Coffee...192`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...193`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...194`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...195`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...196`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...197`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...198`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...199`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...200`     <chr> "Mocha (Without Whipped Cream…
## $ `Frappuccino Blended Coffee...201`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...202`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...203`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...204`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...205`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...206`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...207`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...208`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...209`     <chr> "Caramel (Without Whipped Cre…
## $ `Frappuccino Blended Coffee...210`     <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...211`     <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...212`     <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...213`     <chr> "Java Chip (Without Whipped C…
```

```
## $ `Frappuccino Blended Coffee...214`       <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...215`       <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...216`       <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...217`       <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Blended Coffee...218`       <chr> "Java Chip (Without Whipped C…
## $ `Frappuccino Light Blended Coffee...219` <chr> "Coffee", "Tall Nonfat Milk",…
## $ `Frappuccino Light Blended Coffee...220` <chr> "Coffee", "Grande Nonfat Milk…
## $ `Frappuccino Light Blended Coffee...221` <chr> "Coffee", "Venti Nonfat Milk"…
## $ `Frappuccino Light Blended Coffee...222` <chr> "Mocha", "Tall Nonfat Milk", …
## $ `Frappuccino Light Blended Coffee...223` <chr> "Mocha", "Grande Nonfat Milk"…
## $ `Frappuccino Light Blended Coffee...224` <chr> "Mocha", "Venti Nonfat Milk",…
## $ `Frappuccino Light Blended Coffee...225` <chr> "Caramel", "Tall Nonfat Milk"…
## $ `Frappuccino Light Blended Coffee...226` <chr> "Caramel", "Grande Nonfat Mil…
## $ `Frappuccino Light Blended Coffee...227` <chr> "Caramel", "Venti Nonfat Milk…
## $ `Frappuccino Light Blended Coffee...228` <chr> "Java Chip", "Tall Nonfat Mil…
## $ `Frappuccino Light Blended Coffee...229` <chr> "Java Chip", "Grande Nonfat M…
## $ `Frappuccino Light Blended Coffee...230` <chr> "Java Chip", "Venti Nonfat Mi…
## $ `Frappuccino Blended Creme...231`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...232`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...233`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...234`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...235`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...236`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...237`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...238`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...239`        <chr> "Strawberries & Creme (Withou…
## $ `Frappuccino Blended Creme...240`        <chr> "Vanilla Bean (Without Whippe…
## $ `Frappuccino Blended Creme...241`        <chr> "Vanilla Bean (Without Whippe…
## $ `Frappuccino Blended Creme...242`        <chr> "Vanilla Bean (Without Whippe…
## $ `Frappuccino Blended Creme...243`        <chr> "Vanilla Bean (Without Whippe…

#b
# Data is transposed (observations are in columns, variables in rows).

#c
coffee <- coffee %>%
  pivot_longer(!Beverage_category, names_to = "Beverage_type",
               values_to = "value") %>%
  pivot_wider(names_from = Beverage_category,
              values_from = value)

#d
#coffee <- coffee %>%
#  rename(Beverage_category=Beverage_type)
coffee <- coffee %>%
  separate_wider_delim(Beverage_type, "...", names=c("Beverage_category",
"mess"))
coffee <- coffee%>% select(-mess)
#Alternatively, you can use gsub and rename
```

```r
#e
unique(coffee$`Total Fat (g)`)

##  [1] "0.1" "3.5" "2.5" "0.2" "6"   "4.5" "0.3" "7"   "5"   "0.4" "9"   "1.5"
## [13] "4"   "2"   "8"   "3"   "11"  "0"   "1"   "10"  "15"  "13"  "0.5" "3 2"

# A value of 3 2 instead of 3.2
unique(coffee$`Caffeine (mg)`)

##  [1] "175"    "260"    "330"    "410"    "75"     "150"    "85"     "95"
##  [9] "180"    "225"    "300"    "10"     "20"     "25"     "30"     "0"
## [17] "Varies" "50"     "70"     "120"    "55"     "80"     "110"    "varies"
## [25] "165"    "235"    "90"     NA       "125"    "170"    "15"     "130"
## [33] "140"    "100"    "145"    "65"     "105"

# Values like varies and Varies are not numbers.

#f
unique(coffee$`Total Fat (g)`)

##  [1] "0.1" "3.5" "2.5" "0.2" "6"   "4.5" "0.3" "7"   "5"   "0.4" "9"   "1.5"
## [13] "4"   "2"   "8"   "3"   "11"  "0"   "1"   "10"  "15"  "13"  "0.5" "3 2"

coffee <- coffee %>%
  mutate(`Total Fat (g)`=ifelse(`Total Fat (g)`=="3 2", "3.2", `Total Fat
(g)`))
unique(coffee$`Caffeine (mg)`)

##  [1] "175"    "260"    "330"    "410"    "75"     "150"    "85"     "95"
##  [9] "180"    "225"    "300"    "10"     "20"     "25"     "30"     "0"
## [17] "Varies" "50"     "70"     "120"    "55"     "80"     "110"    "varies"
## [25] "165"    "235"    "90"     NA       "125"    "170"    "15"     "130"
## [33] "140"    "100"    "145"    "65"     "105"

coffee <- coffee %>%
  mutate(`Caffeine (mg)`=ifelse(`Caffeine (mg)`=="Varies", NA, `Caffeine
(mg)`)) %>%
  mutate(`Caffeine (mg)`=ifelse(`Caffeine (mg)`=="varies", NA, `Caffeine
(mg)`))
#You can also find a solution to solve with ignoring the case

#g
coffee[4:18] <- sapply(coffee[4:18],as.numeric)
coffee <- coffee %>% mutate(across(4:18, as.numeric))

#h
cafeine <- max(coffee$`Caffeine (mg)`, na.rm=TRUE)
coffee$`Sugars (g)`[coffee$`Caffeine (mg)`==cafeine & !is.na(coffee$`Caffeine
(mg)`)]

## [1] 0
```

```r
#Exercise 5.2
# Propose a solution for the NAs using the information in the dataset only.
# Implement your proposal.

iris <- read_csv("dirty petal.csv")

summary(iris)

##   Sepal.Length     Sepal.Width      Petal.Length    Petal.Width
##  Min.   : 0.000   Min.   :-3.000   Min.   : 0.00   Min.   :0.1
##  1st Qu.: 5.100   1st Qu.: 2.800   1st Qu.: 1.60   1st Qu.:0.3
##  Median : 5.750   Median : 3.000   Median : 4.50   Median :1.3
##  Mean   : 6.559   Mean   : 3.391   Mean   : 4.45   Mean   :Inf
##  3rd Qu.: 6.400   3rd Qu.: 3.300   3rd Qu.: 5.10   3rd Qu.:1.8
##  Max.   :73.000   Max.   :30.000   Max.   :63.00   Max.   :Inf
##  NA's   :10       NA's   :17       NA's   :19      NA's   :12
##    Species
##  Length:150
##  Class :character
##  Mode  :character
##


# Are they in the same rows or not?
filter(iris, is.na(Sepal.Length))

## # A tibble: 10 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1           NA         3.9         1.7          0.4 setosa
##  2           NA         4          NA            0.2 setosa
##  3           NA         3           5.9          2.1 virginica
##  4           NA         2.8         0.82         1.3 versicolor
##  5           NA         2.9         4.5          1.5 versicolor
##  6           NA         3.2         5.7          2.3 virginica
##  7           NA         3.3         5.7          2.1 virginica
##  8           NA         3           5.5          2.1 virginica
##  9           NA         2.8         4.7          1.2 versicolor
## 10           NA         3           4.9          1.8 virginica

filter(iris, is.na(Sepal.Width))

## # A tibble: 17 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          6.2          NA        5.4           2.3 virginica
##  2          5.3          NA       NA             0.2 setosa
##  3          5.5          NA        4             1.3 versicolor
##  4          5.6          NA        4.2           1.3 versicolor
##  5          5.4          NA        4.5           1.5 versicolor
##  6          5            NA        1.2           0.2 setosa
##  7          0            NA        1.3           0.4 setosa
##  8          5.7          NA       NA             0.4 setosa
##  9          6.5          NA        4.6           1.5 versicolor
## 10          4.9          NA        3.3           1   versicolor
```

```
## 11          6.7          NA        5              1.7 versicolor
## 12          5.5          NA        0.925          1   versicolor
## 13          6.3          NA        4.4            1.3 versicolor
## 14          5.8          NA        5.1            2.4 virginica
## 15          4.8          NA        1.9            0.2 setosa
## 16          5            NA        1.4            0.2 setosa
## 17          5.5          NA        3.8            1.1 versicolor
```

```
filter(iris, is.na(Petal.Length))
```

```
## # A tibble: 19 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          5.3        NA             NA         0.2 setosa
##  2         NA           4             NA         0.2 setosa
##  3          4.9         3.6           NA         0.1 setosa
##  4          6.2         2.8           NA         1.8 virginica
##  5          4.4         3.2           NA         0.2 setosa
##  6          6.2         2.9           NA         1.3 versicolor
##  7          5.7        NA             NA         0.4 setosa
##  8          4.9         3.1           NA         0.2 setosa
##  9          5.1         2.5           NA         1.1 versicolor
## 10          5.1         3.5           NA        NA   setosa
## 11          6.9         3.1           NA         1.5 versicolor
## 12          7.2         3.6           NA         2.5 virginica
## 13          5.1         3.7           NA         0.4 setosa
## 14          6.1         2.8           NA         1.3 versicolor
## 15          6.3         3.4           NA         2.4 virginica
## 16          5.9         3             NA         1.5 versicolor
## 17          4.6         3.6           NA         0.2 setosa
## 18          7.7         3             NA         2.3 virginica
## 19          6.4         3.1           NA         1.8 virginica
```

```
filter(iris, is.na(Petal.Width))
```

```
## # A tibble: 12 × 5
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##           <dbl>       <dbl>        <dbl>       <dbl> <chr>
##  1          6.4         2.7          5.3          NA virginica
##  2          6           3            4.8          NA virginica
##  3         73          29           63            NA virginica
##  4          5.9         3.2          4.8          NA versicolor
##  5          6.4         2.8          5.6          NA virginica
##  6          5           2.3          3.3          NA versicolor
##  7          5.1         3.5         NA            NA setosa
##  8          4.9         2.5          4.5          NA virginica
##  9          6.3         2.7          4.9          NA virginica
## 10          4.8         3            1.4          NA setosa
## 11          4.4         3            1.3          NA setosa
## 12          5.8         2.6          4            NA versicolor
```

```
# As no NA in the Species column, we can use this information.
# A possible strategy:
# - delete rows where there are two missing values
# - replace the only missing value by the mean of the species
```

```r
iris$missing <- is.na(iris$Sepal.Length)+
  is.na(iris$Sepal.Width)+
  is.na(iris$Petal.Length)+
  is.na(iris$Petal.Width)

iris <- iris %>% filter(missing<2)

#To check if it is OK.
summary(iris$missing)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  0.0000  0.3425  1.0000  1.0000
```

```r
# Replace missing values with the mean of the column for the given species
cleaned_iris <- iris %>%
  group_by(Species) %>% # Group by Species
  mutate(across(everything(), ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)))
%>%
  select(!missing)

# you can also calculate the mean of each group and species
# and use it in 12 mutates
# mean(iris$Sepal.Length[iris$Species=="setosa"], na.rm = TRUE)
```